

Same Score, Different Standing

AP Statistics · Unit 2 · Topic 2.11 · B: 2026-11-10 · E: 2026-12-04 · TEACHER KEY

CED TETHER

- SKILLS: Using Probability and Simulation
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 3.C** — Describe probability distributions.
- ENDURING UNDERSTANDING: VAR-6 The normal distribution may be used to model variation.
- VAR-6.A Calculate the probability that a particular value lies in a given interval of a normal distribution. [Skill 3.A]
- VAR-6.A.1 A continuous random variable is a variable that can take on any value within a specified domain. Every interval within the domain has a probability associated with it. VAR-6.A.2 A continuous random variable with a normal distribution is commonly used to describe populations. The distribution of a normal random variable can be described by a normal, or "bell-shaped," curve. VAR-6.A.3 The area under a normal curve over a given interval represents the probability that a particular value lies in that interval.
- VAR-6.B Determine the interval associated with a given area in a normal distribution. [Skill 3.A]
- VAR-6.B.1 The boundaries of an interval associated with a given area in a normal distribution can be determined using z-scores or technology.
- VAR-6.C Determine the appropriateness of using the normal distribution to approximate probabilities for unknown distributions. [Skill 3.C]
- VAR-6.C.1 Normal distributions are symmetrical and "bell-shaped." As a result, normal distributions can be used to approximate distributions with similar characteristics.

Essential question: How can the same raw score have different relative standings in two normal distributions with the same mean?

DOK-3 skill: 4.B · **FRQ pattern:** same-score-two-models-compare-with-z

DOK rationale: Part (c) coordinates two standardized scores and normal areas to adjudicate a same-percentile claim, explain the role of spread, and define what a relative comparison can support.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask for a claim before students calculate either percentile.
- Begin the combined 5.1–5.2 video follow-along at minute 5.
- Return to the board slide at minute 33. Circulate and require both z-scores before a comparison.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose whether the percentile is the same and cite mean or standard deviation.
- Complete the follow-along about sampling variability, normal probabilities, and normal-model appropriateness.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- How many School A standard deviations above 80 is a 92?
- Why does School B's wider curve place more area above the same score?
- Which area gives a percentile: left of 92 or right of 92?
- What exactly do you mean by 'better' when the raw score is identical?

Adult role

Solo teacher. Circulate 5 pairs. Require students to translate raw-point distance into standard-deviation units and define better as relative standing before choosing a school.

[WARN] WATCH FOR

- Using the shared mean as evidence that all corresponding percentiles match.
- Reporting the upper-tail probability rather than the percentile below 92.
- Reversing the comparison and assigning the higher percentile to the wider distribution.

SCORING PART (c) — E / P / I

Expected elements

- **rejects-same-percentile** (required) — Rejects the claim that a score of 92 has the same percentile in the two models
- **uses-two-standardizations** (required) — Cites z approximately 1.33 and percentile 90.9 percent for School A, and z 0.80 and percentile 78.8 percent for School B
- **explains-spread-mechanism** (required) — Explains that the same 12-point distance represents different numbers of standard deviations because the standard deviations differ
- **makes-relative-choice** (required) — Chooses School A for the higher relative standing and limits better to the stated percentile meaning
- **states-reader-consequence** (required) — Explains that the counselor's claim hides the effect of variability on standardized position

E	Rejects the claim, uses both correct z -scores and percentiles, links the difference to standard deviation, identifies School A as the higher relative standing, and explains the misleading implication.
P	Reaches the correct school and gives both percentiles, but incompletely explains the standard-deviation mechanism, the meaning of better, or the consequence of the claim.
I	Says equal means force equal percentiles, compares only raw distances, or reverses which school gives 92 the higher percentile.

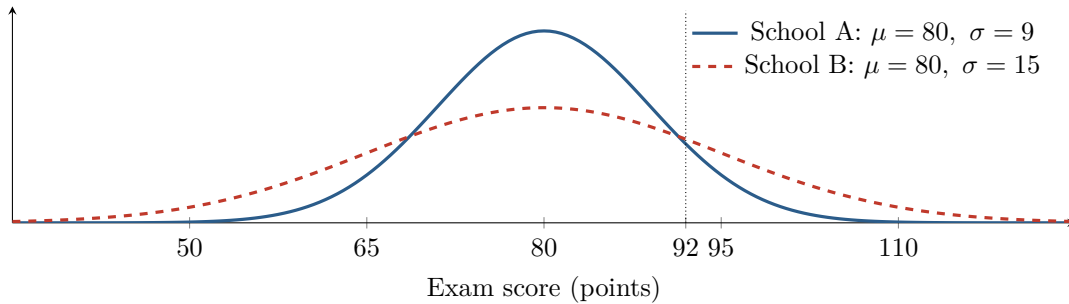
Common mistakes

- Using 80 as the standard deviation because it is shared by both models
- Computing $z = (80 - 92)/\sigma$ and reporting lower-tail areas without correcting direction
- Saying the wider distribution makes every above-mean score more exceptional
- Calling one raw score objectively higher even though the numerical score is 92 at both schools

[STAR] TODAY'S PROBLEM — annotated

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE (5 min)

Is a score of 92 at the same percentile in both schools? Choose a claim and name one parameter behind your choice.

Key: Not graded. Equal means should make the counselor’s claim tempting; standardization should change the final judgment.

Rules callout “STANDARDIZE BEFORE COMPARING (keep on the page)” is printed on the student sheet.

DOK 1 (a) Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

Key: School B is more variable because $\sigma_B = 15$ points is larger than $\sigma_A = 9$ points. Its normal curve is wider and lower; School A’s is narrower and taller around the common mean of 80.

DOK 2 (b) For a score of 92, calculate the z -score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

Key: School A: $z = (92 - 80)/9 = 1.333\dots$, so $P(X_A \leq 92) = \Phi(1.333\dots) \approx 0.9088$ and 92 is at about the 90.9th percentile. School B: $z = (92 - 80)/15 = 0.80$, so $P(X_B \leq 92) = \Phi(0.80) \approx 0.7881$ and 92 is at about the 78.8th percentile.

DOK 3 (c) Decide whether the counselor’s same-percentile claim is warranted. Cite both z -scores and percentiles, explain why equal means do not give equal relative standing when standard deviations differ, and state at which school a 92 is ‘better’ if better means a higher standing relative to that school’s scores. Explain what the counselor’s claim would mislead a reader into believing.

Key: The counselor’s claim is not warranted. Although 92 is 12 points above the common mean, that gap is 1.33 School A standard deviations but only 0.80 School B standard deviations. The corresponding percentiles are about 90.9 and 78.8, not the same. If ‘better’ means higher relative standing, the 92 is better at School A. Saying the percentiles are equal would mislead a reader into treating distance in raw points as distance in standard-deviation units and into overlooking School B’s greater variability.

EXIT

One thing the video changed about my first take: _____